

A3-PS018-TC360

HAPS Link-Quality Prediction Using a Documented Channel Model

Mahesh N
Mahesh.mnraj@gmail.com

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Problem Description

HAPS communication systems provide a potential means of extending broadband coverage beyond conventional terrestrial infrastructure. However, received signal quality depends on propagation loss, link geometry, and environmental attenuation.

Key Issues:

- Estimating propagation loss for a HAPS-to-ground communication link.
- Calculating received signal power and Signal-to-Noise Ratio (SNR) using a documented link budget.
- Generating a reproducible synthetic dataset under two environmental scenarios.
- Predicting link quality for previously unseen receiver trajectories.
- Quantifying prediction error against an analytical baseline.
- Identifying the influence of propagation and link-budget parameters on communication performance.

Background

- High-Altitude Platform Stations (HAPS) operate in the stratosphere and can support non-terrestrial broadband connectivity over large geographical areas. Unlike conventional terrestrial base stations, HAPS operate at a much greater altitude. Therefore, communication link geometry and propagation conditions must be considered when estimating received signal quality.
- The received signal power depends on transmitted power, transmitting and receiving antenna gains, and total propagation loss. The link budget is expressed as:

$$P_r = P_t + G_t + G_r - L_{total}$$

- where all power and loss quantities are expressed in decibel-compatible units.
- Signal-to-noise ratio is calculated from the received signal power and receiver noise power:

$$SNR_{(dB)} = P_r - N_{(dBm)}$$

- A documented propagation model provides the analytical basis for estimating path loss. ITU-R P.528-5 is a relevant candidate because it provides a propagation prediction method for aeronautical services within a specified frequency range. However, its scope and limitations must be considered carefully when applying it to a HAPS scenario.
- The project combines propagation modelling, link-budget calculations, synthetic dataset generation, prediction, and quantitative evaluation [1].
- **Reference [1]:** International Telecommunication Union, *Recommendation ITU-R P.528-5: A propagation prediction method for aeronautical mobile and radionavigation services using the VHF, UHF and SHF bands*, 2021.

Literature & RFC Review

1. ITU-R P.528-5: Propagation Prediction for Aeronautical Services

ITU-R P.528-5 specifies a method for predicting basic transmission loss for aeronautical mobile and radionavigation services over the 100 MHz–30 GHz frequency range. It provides an analytical foundation for estimating propagation loss using parameters such as terminal heights, distance, frequency, polarization, and time percentage [1].

2. ITU-R P.838: Specific Attenuation Model for Rain

ITU-R P.838 can be used to calculate specific attenuation due to rain when implementing a rain-affected scenario. The calculation requires documented assumptions about frequency, rainfall rate, polarization, and effective path length [2].

3. Link-Budget Analysis

Link-budget calculations combine transmit power, antenna gains, and propagation losses to estimate received signal power. Receiver bandwidth and noise assumptions are then used to calculate Signal-to-Noise Ratio (SNR).

4. Synthetic Dataset-Based Prediction

Synthetic data generated from a documented analytical model can be used to train and evaluate a lightweight prediction model. However, prediction accuracy measures how well the model reproduces the analytical baseline and does not independently establish real-world communication performance

Research Gap Addressed

The project integrates a documented propagation model, reproducible dataset generation, environmental scenario comparison, held-out trajectory evaluation, and sensitivity analysis into one compact HAPS simulation framework.

Proposed Solution

The proposed system estimates HAPS communication link quality using a documented propagation model, analytical link-budget calculations, and a reproducible synthetic dataset.
The implementation is divided into the following modules.

1. HAPS Geometry Definition

A HAPS is positioned at a fixed altitude of 20 km. Ground-receiver trajectories are defined to represent different communication-link geometries.
The operating frequency is kept constant throughout the experiment.

2. Propagation-Loss Calculation

A documented propagation model is used to calculate the analytical path-loss baseline for each valid link configuration.
The model inputs, units, assumptions and applicability are recorded.

3. Link-Budget Calculation

The propagation-loss results are combined with transmit power and antenna gains to calculate received signal power.
SNR is calculated using the specified receiver bandwidth and noise assumptions.

4. Environmental Scenario Generation

Two environmental scenarios are implemented using clearly defined assumptions. Any additional attenuation terms are calculated using documented models rather than arbitrary values.

5. Synthetic Dataset Generation

The system generates reproducible samples containing trajectory information, link geometry, propagation loss, received power, SNR and environmental scenario.

6. Prediction and Evaluation

A lightweight prediction model is trained using designated trajectories. Separate trajectories are reserved for testing.
The model's predictions are compared with analytical baseline values using MAE.

7. Visualization and Demonstration

Graphs illustrate path loss, received power, scenario differences, prediction errors and sensitivity to selected parameters.

Optimization Proposed

Aspect	Initial Approach (Before)	Proposed Improvement (After)
Propagation Modelling	Unstructured path-loss calculations	Documented propagation model with explicit assumptions
Link-Budget Calculation	Individual calculations without a common configuration	Standardized calculation module with consistent units
Dataset Generation	Manually entered samples	Automated and reproducible dataset generator
Environmental Analysis	Single-condition analysis	Comparison of two defined environmental scenarios
Evaluation	Training and testing on potentially overlapping samples	Held-out trajectory evaluation
Prediction Assessment	Qualitative comparison	Quantitative MAE calculation
Parameter Analysis	Fixed assumptions without systematic testing	Sensitivity analysis of selected parameters
Demonstration	Separate scripts and plots	Integrated execution workflow

Solution Architecture & Design

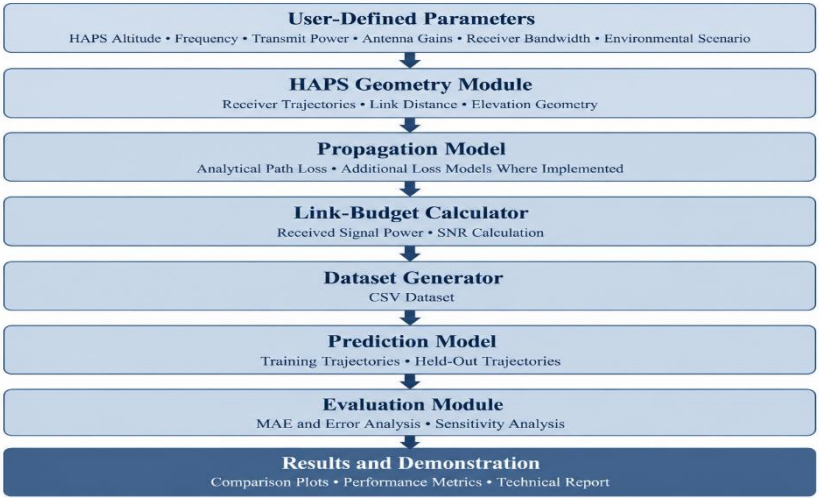


Figure: System Architecture of the proposed HAPS link-quality prediction framework

Data Flow Diagram

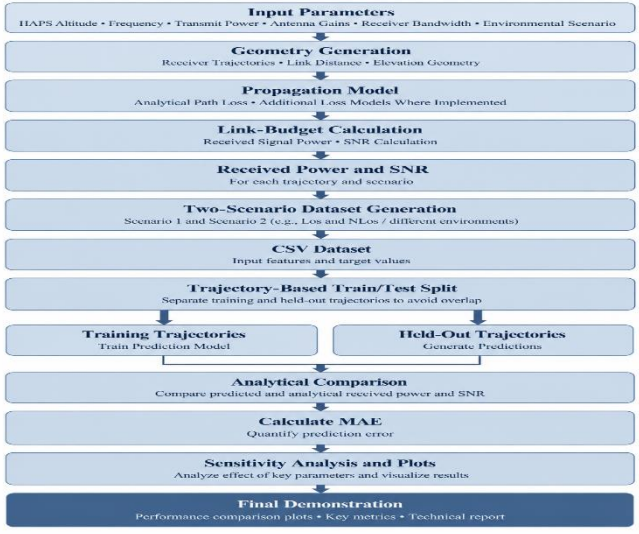


Figure: Data Flow Diagram of the HAPS link-quality prediction system

Deployment Timeline

Phase	Timeline	Key Activities	Deliverable
Phase 1: Foundation	16 – 20 October	• Finalize assumptions • Implement propagation model • Define geometry and calculate link budget	Validated analytical baseline
Phase 2: Dataset and Prediction	21 – 24 October	• Generate dataset • Split trajectories • Implement prediction model • Calculate MAE	Working end-to-end prototype
Phase 3: Optimization	25 – 28 October	• Implement environmental scenarios • Perform sensitivity analysis • Validate reproducibility • Build demonstration	Evaluated and demonstrable prototype
Phase 4: Documentation and Submission	29 – 31 October	• Complete report • Demonstrate prototype • Finalize GitHub • Verify deliverables	Final technical report and demonstration

Development Methodology

Incremental implementation and testing of individual components, followed by integration, analytical baseline validation, dataset generation, held-out trajectory evaluation, sensitivity analysis, documentation and final demonstration

Bibliography I

[1] International Telecommunication Union (ITU-R), *Recommendation ITU-R P.528-5: A propagation prediction method for aeronautical mobile and radionavigation services using the VHF, UHF and SHF bands*, 2021.
[2] International Telecommunication Union (ITU-R), *Recommendation ITU-R P.838: Specific attenuation model for rain for use in prediction methods*.
[3] International Telecommunication Union (ITU-R), *Recommendation ITU-R P.525: Calculation of free-space attenuation*.
[4] International Telecommunication Union (ITU-R), *Recommendation ITU-R P.618: Propagation data and prediction methods required for the design of Earth-space telecommunication systems*.

Thank You for Your Attention!